

Test Plan

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System

Type: Hotel Reservation

Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)

Date: February 20, 2026

Version: 1.0

1. Test Plan Overview

1.1 Purpose

This test plan defines the strategy, scope, and approach for testing the StayEase Hotel Reservation System. The goal is to verify that all functional and non-functional requirements are met, reservation workflows operate correctly without double-booking errors, and the system is ready for production deployment at Grand Vista Hotel.

1.2 Scope

In Scope

- All functional requirements (FR-1 through FR-12)
- Critical non-functional requirements (performance, security, usability, scalability)
- Core user workflows: registration, room search, booking, modification, cancellation, check-in/check-out
- Staff dashboard functionality and reporting

Out of Scope

- Third-party email delivery reliability (SendGrid SLA)
- Infrastructure and hardware testing
- Load testing beyond 100 concurrent users
- Mobile browser compatibility (V1 is desktop-first)

2. Test Strategy

2.1 Testing Levels

Unit Testing

- Scope: Individual functions, validators, date overlap logic, price calculators
- Who: Developers (Kyrylo, Danylo)
- When: During implementation phase
- Coverage Goal: 80% code coverage for business logic modules

Integration Testing

- Scope: API endpoint behavior, database queries, frontend-backend data flow
- Who: QA Lead (Daniil) and Developers
- When: After unit testing, before system testing
- Focus: REST API correctness, reservation transaction integrity

System Testing

- Scope: Complete end-to-end reservation workflows
- Who: QA Lead (Daniil)
- When: After integration testing
- Focus: User scenarios from registration through checkout

User Acceptance Testing (UAT)

- Scope: Real-world usage by hotel manager and selected staff
- Who: Elena Petrova (Hotel Manager), front desk staff
- When: Final phase before deployment
- Focus: Meets business needs, usability, operational readiness

2.2 Testing Approach

- Manual Testing: UI workflows, usability evaluation, visual verification
- Automated Testing: Unit tests (Jest), API tests (Supertest), regression suite
- Exploratory Testing: Ad-hoc testing for edge cases (overlapping dates, boundary conditions, concurrent bookings)

3. Test Cases

3.1 Functional Test Cases

TC-01: Guest Login with Valid Credentials

Requirement: FR-2 (Guest Registration and Login)

Priority: High

Preconditions: Guest account exists (email: guest@test.com, password: Test123!)

Test Steps:

1. Navigate to login page
2. Enter email: guest@test.com
3. Enter password: Test123!
4. Click 'Login' button

Expected Result: Guest is redirected to room search page. Guest name appears in header. JWT token stored in browser.

Status: [Not Tested]

TC-02: Guest Login with Invalid Password

Requirement: FR-2

Priority: High

Preconditions: Guest account exists

Test Steps:

1. Navigate to login page
2. Enter email: guest@test.com
3. Enter incorrect password: WrongPass
4. Click 'Login'

Expected Result: Login fails. Error message: 'Invalid email or password'. User stays on login page.

Status: [Not Tested]

TC-03: Display Room Catalog

Requirement: FR-1 (Display Room Catalog)

Priority: High

Preconditions: Database contains at least 4 room types with photos and pricing

Test Steps:

- 1. Log in as guest
- 2. Navigate to room catalog page

Expected Result: All room types displayed with name, photo, description, amenities, capacity, and nightly rate.

Status: [Not Tested]

TC-04: Search Available Rooms

Requirement: FR-3 (Search Available Rooms)

Priority: High

Preconditions: Rooms exist with varying availability for March 2026

Test Steps:

- 1. Log in as guest
- 2. Set check-in: March 15, 2026
- 3. Set check-out: March 18, 2026
- 4. Set guests: 2
- 5. Click 'Search'

Expected Result: Only rooms available for the entire March 15-18 range are shown. Fully booked rooms are excluded. Each result shows room type, price per night, total price for 3 nights.

Status: [Not Tested]

TC-05: Make Reservation Successfully

Requirement: FR-4 (Make Reservation)

Priority: High

Preconditions: Guest logged in. Deluxe Room 201 is available for selected dates.

Test Steps:

- 1. Search rooms for March 15-18
- 2. Select 'Deluxe Room'
- 3. Click 'Book Now'
- 4. Enter special requests: 'Late check-in, around 10 PM'
- 5. Click 'Confirm Booking'

Expected Result: Booking confirmed. Confirmation page shows reservation ID, room details, dates, total price. Room availability updates immediately. Confirmation email sent.

Status: [Not Tested]

TC-06: Prevent Double Booking

Requirement: FR-5 (Prevent Double Booking)

Priority: High

Preconditions: Room 201 is already booked for March 15-18.

Test Steps:

- 1. Log in as a different guest
- 2. Search rooms for March 16-17 (overlapping dates)
- 3. Attempt to book Room 201

Expected Result: Room 201 does not appear in search results. If accessed directly via URL, booking attempt fails with 'Room unavailable for selected dates' message. System suggests alternative rooms.

Status: [Not Tested]

TC-07: Cancel Reservation (More than 24 Hours Before Check-In)

Requirement: FR-6 (Cancel Reservation)

Priority: High

Preconditions: Guest has reservation for March 15-18. Current date is March 13.

Test Steps:

- 1. Navigate to 'My Reservations'
- 2. Find March 15-18 reservation
- 3. Click 'Cancel'
- 4. Confirm cancellation

Expected Result: Reservation cancelled. Status changes to 'Cancelled'. Room becomes available for those dates again. Cancellation email sent.

Status: [Not Tested]

TC-08: Prevent Cancellation Within 24 Hours of Check-In

Requirement: FR-6

Priority: Medium

Preconditions: Guest has reservation checking in tomorrow. Less than 24 hours remain.

Test Steps:

- 1. Navigate to 'My Reservations'
- 2. Find the reservation
- 3. Attempt to click 'Cancel'

Expected Result: Cancel button is disabled. Message: 'Cannot cancel within 24 hours of check-in. Please contact front desk.'

Status: [Not Tested]

TC-09: Modify Reservation Dates

Requirement: FR-7 (Modify Reservation)

Priority: Medium

Preconditions: Guest has reservation for March 15-18. Room is available for March 15-20.

Test Steps:

- 1. Navigate to 'My Reservations'

- 2. Click 'Modify' on March 15-18 reservation
- 3. Change check-out to March 20
- 4. Confirm modification

Expected Result: Reservation updated. New dates and updated total price displayed. Modification confirmation email sent.

Status: [Not Tested]

TC-10: View Reservation History

Requirement: FR-8 (View Reservation History)

Priority: Medium

Preconditions: Guest has 3 reservations (1 upcoming, 1 completed, 1 cancelled)

Test Steps:

- 1. Navigate to 'My Reservations'

Expected Result: All 3 reservations displayed with room type, dates, total price, and status badges (Confirmed, Completed, Cancelled). Sorted by date, upcoming first.

Status: [Not Tested]

TC-11: Staff Check-In Guest

Requirement: FR-10 (Check-In and Check-Out)

Priority: Low

Preconditions: Staff logged in. Guest has confirmed reservation for today.

Test Steps:

- 1. Open Staff Dashboard
- 2. Find guest in 'Today's Check-Ins'
- 3. Click 'Check In'

Expected Result: Reservation status changes to 'Checked In'. Room status changes to 'Occupied'. Guest removed from check-in list.

Status: [Not Tested]

TC-12: Staff Dashboard Overview

Requirement: FR-9 (Front Desk Dashboard)

Priority: Medium

Preconditions: Staff logged in. Multiple reservations exist for today.

Test Steps:

- 1. Log in as staff
- 2. View dashboard

Expected Result: Dashboard shows today's check-ins count, today's check-outs count, current occupancy rate, and searchable reservation list.

Status: [Not Tested]

3.2 Non-Functional Test Cases

TC-13: Performance - Room Search Response Time

Requirement: NFR-1 (Performance)

Priority: High

Preconditions: Database seeded with 50 rooms and 500 reservations

Test Steps:

- 1. Log in as guest
- 2. Enter search criteria
- 3. Click Search
- 4. Measure time from click to results displayed (using browser DevTools)

Expected Result: Search results appear within 2 seconds. Booking creation completes within 3 seconds.

Status: [Not Tested]

TC-14: Usability - First-Time Guest Books Room

Requirement: NFR-2 (Usability)

Priority: High

Preconditions: New user, never used the system

Test Steps:

- 1. Give user the task: 'Book a Deluxe room for March 15-18 for 2 guests'
- 2. Observe without providing help
- 3. Measure time to complete
- 4. Note any confusion or errors

Expected Result: User completes booking in under 3 minutes. No help needed. User reports task as 'easy'. Process requires 5 or fewer steps.

Status: [Not Tested]

TC-15: Security - Password Hashing Verification

Requirement: NFR-4 (Security)

Priority: High

Preconditions: New user account created

Test Steps:

- 1. Create a new guest account with password: SecurePass123!
- 2. Query database: `SELECT password_hash FROM users WHERE email = 'test@test.com'`
- 3. Verify password is hashed

Expected Result: Password is NOT stored in plaintext. Hash starts with '\$2a\$' or '\$2b\$' (bcrypt format). Hash is 60 characters long.

Status: [Not Tested]

4. Requirements Traceability Matrix

Requirement	Description	Test Cases	Status
FR-1	Display Room Catalog	TC-03	[Not Tested]
FR-2	Guest Registration/Login	TC-01, TC-02	[Not Tested]
FR-3	Search Available Rooms	TC-04	[Not Tested]
FR-4	Make Reservation	TC-05	[Not Tested]
FR-5	Prevent Double Booking	TC-06	[Not Tested]
FR-6	Cancel Reservation	TC-07, TC-08	[Not Tested]
FR-7	Modify Reservation	TC-09	[Not Tested]
FR-8	View Reservation History	TC-10	[Not Tested]
FR-9	Front Desk Dashboard	TC-12	[Not Tested]
FR-10	Check-In / Check-Out	TC-11	[Not Tested]
FR-11	Email Notifications	TC-05 (partial)	[Not Tested]
FR-12	Occupancy Reporting	[Add TC]	[Not Tested]
NFR-1	Performance	TC-13	[Not Tested]
NFR-2	Usability	TC-14	[Not Tested]
NFR-3	Reliability	[Add TC]	[Not Tested]
NFR-4	Security	TC-15	[Not Tested]
NFR-5	Scalability	[Add TC]	[Not Tested]

Coverage: 14/17 requirements have dedicated test cases.

5. Test Environment

5.1 Hardware

- Test Server: DigitalOcean Droplet (2 vCPU, 4GB RAM, Ubuntu 22.04)
- Client Devices: Desktop (Chrome, Firefox), Tablet (Safari)

5.2 Software

- Backend: Node.js 20, Express.js 4.x
- Database: PostgreSQL 15
- Browsers: Chrome (latest), Firefox (latest)
- Test Frameworks: Jest (unit), Supertest (API), Cypress (E2E)

5.3 Test Data

- Users: 50 guest accounts, 5 staff accounts, 1 manager account
- Rooms: 50 rooms (20 standard, 15 deluxe, 10 suite, 5 family)
- Reservations: 200 reservations (mix of past, current, future, cancelled)

6. Test Schedule

Testing Phase	Start Date	End Date	Duration	Owner
Unit Testing	Apr 4	Apr 7	4 days	Kyrylo, Danylo
Integration Testing	Apr 8	Apr 10	3 days	Daniil
System Testing	Apr 11	Apr 13	3 days	Daniil
UAT	Apr 14	Apr 16	3 days	Elena Petrova
Bug Fixes	Apr 17	Apr 18	2 days	Kyrylo, Danylo
Regression Testing	Apr 19	Apr 19	1 day	Daniil

7. Entry and Exit Criteria

7.1 Entry Criteria

- ☒ All code implemented and merged to main branch
- ☒ Unit tests written and passing (80% coverage)
- ☒ Test environment set up and accessible
- ☒ Test data seeded in database
- ☒ Test plan reviewed and approved

7.2 Exit Criteria

- ☒ All 15 test cases executed
- ☒ 100% of high-priority requirements tested and passing
- ☒ No critical or high-severity bugs remain open
- ☒ UAT passed by hotel manager
- ☒ Test summary report completed

8. Defect Management

8.1 Bug Severity Levels

Critical: System crash, data loss, security breach, double booking occurs. Action: Fix immediately, halt testing.

High: Core reservation workflow broken, cannot book/cancel. Action: Fix before deployment.

Medium: Feature partially broken, workaround exists (e.g., search filter not working). Action: Fix if time allows.

Low: Minor UI issue, cosmetic problem (misaligned element, typo). Action: Log for future release.

8.2 Bug Reporting Template

Bug ID: [BUG-XXX] | Title: [Short description] | Severity: [Critical/High/Medium/Low]

Steps to Reproduce: [Numbered steps] | Expected: [What should happen] | Actual: [What happened]

Screenshot: [If applicable] | Assigned To: [Developer name]

9. Test Summary Report

(To be filled after testing completes)

9.1 Test Execution Summary

Metric	Count	Percentage
Total Test Cases	15	100%
Executed	[TBD]	[TBD]
Passed	[TBD]	[TBD]
Failed	[TBD]	[TBD]
Not Tested	[TBD]	[TBD]

9.2 Defects Summary

Severity	Found	Fixed	Open
Critical	0	0	0
High	[TBD]	[TBD]	[TBD]
Medium	[TBD]	[TBD]	[TBD]
Low	[TBD]	[TBD]	[TBD]

9.3 Recommendation

[To be completed after testing. Expected: System is ready for deployment once all critical and high-severity bugs are resolved.]

10. Sign-Off

Prepared By: Daniil Ustiuhov (QA Lead)

Reviewed By: Kyrylo Pryiomyshev (PM) / Nicolai Morozov (Instructor)

Approval Date: February 23, 2026

QA Lead Signature: _____ Date: _____

Stakeholder Signature: _____ Date: _____